

PG25 erosion lighting talk

COOLING:
COLD PLATE

Erosion study – High velocity PG25 flow

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OCP PLATINUM® Member

Goal and approach

Reference fluid velocity guidance

Water temperature	Velocity limits
Cold water	< 8 ft/s (2.4 m/s)
Hot water < 140 °F (60 °C)	< 5 ft/s (1.5 m/s)
Hot water > 140 °F (60 °C)	< 2 – 3 ft/s (0.6 – 0.9 m/s)

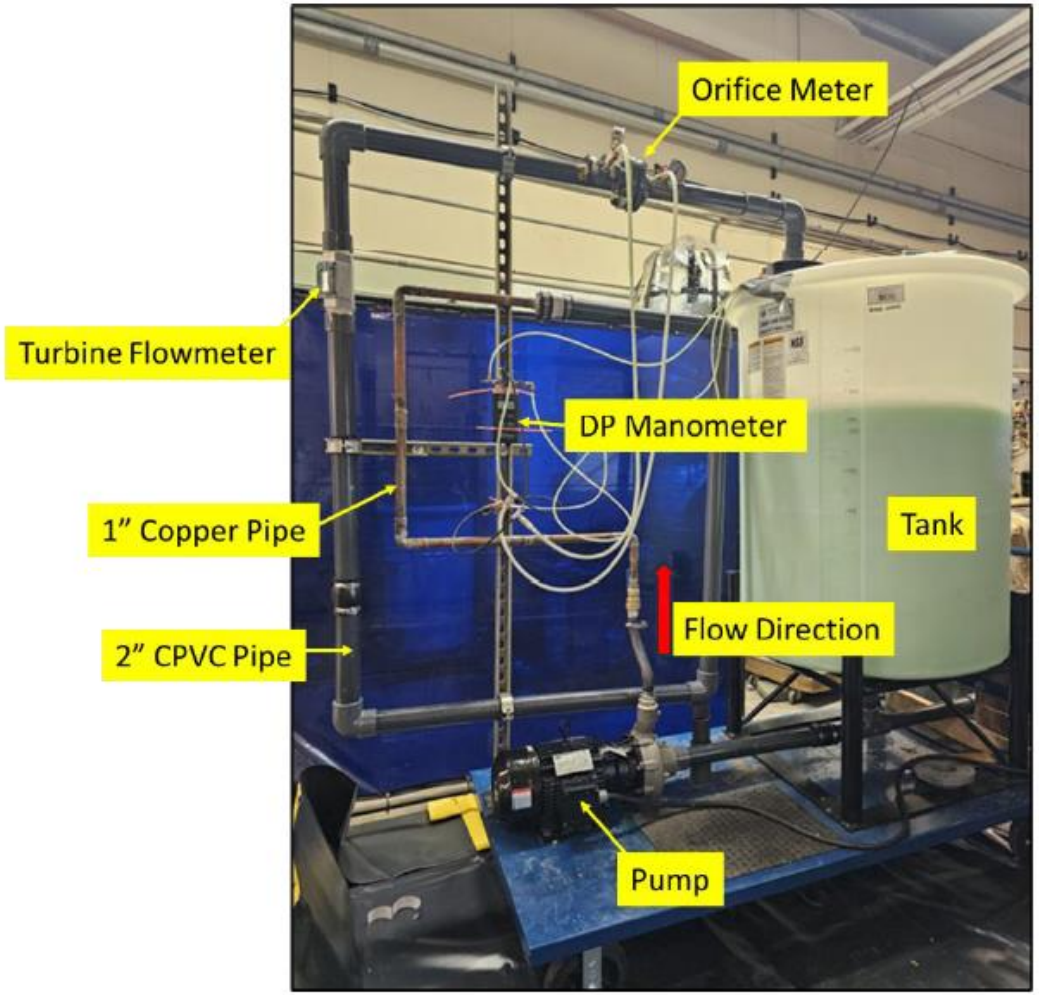
CDA Publication A4015-26/24: Copper Tube Handbook

- Copper Development Association defines water velocity limits for copper tubes due to possibility of erosion-corrosion

- Data center industry needs scientific evidence to understand how PG25 fluids perform under increased flow velocity

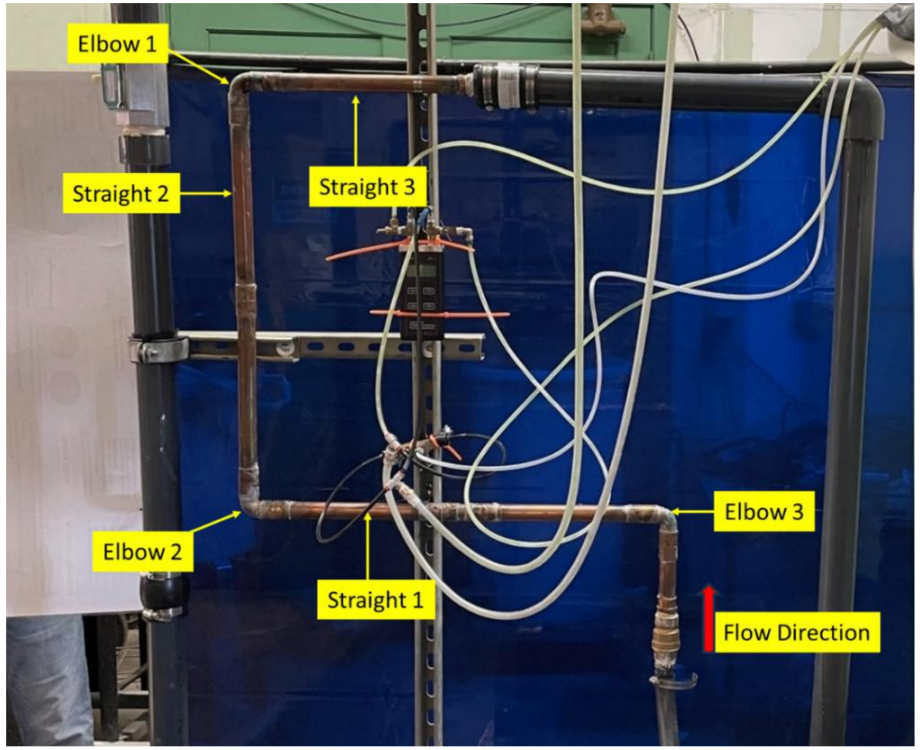
Phase 1	Phase 2
Design and operate the rig. Monitor copper wall thickness and fluid condition Lead: Castrol	Conduct in-depth post-test surface and material analysis via XPS Lead: Meta

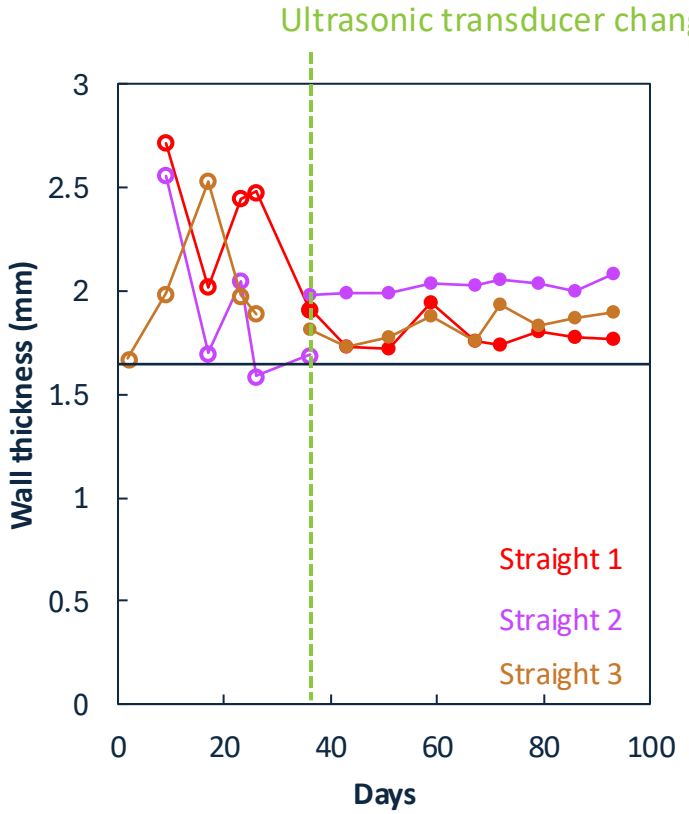
Liquid loop design

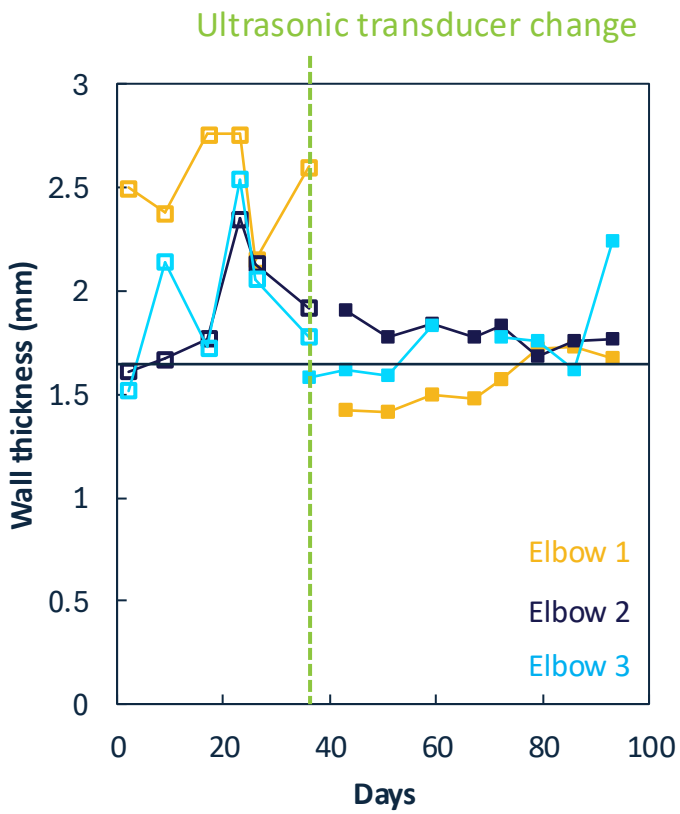


- Fluid: Castrol ON Direct Liquid Cooling PG25
- Fluid velocity: 6.2m/s
- Copper pipe:
  - 25.4mm (1") ID, thickness 1.65mm (0.065")
  - C12200 alloy, H58 temper
  - Non-passivated
- Test duration: 2,232h (93 days)
- Set up at the University of Tulsa

Wall thickness remained constant



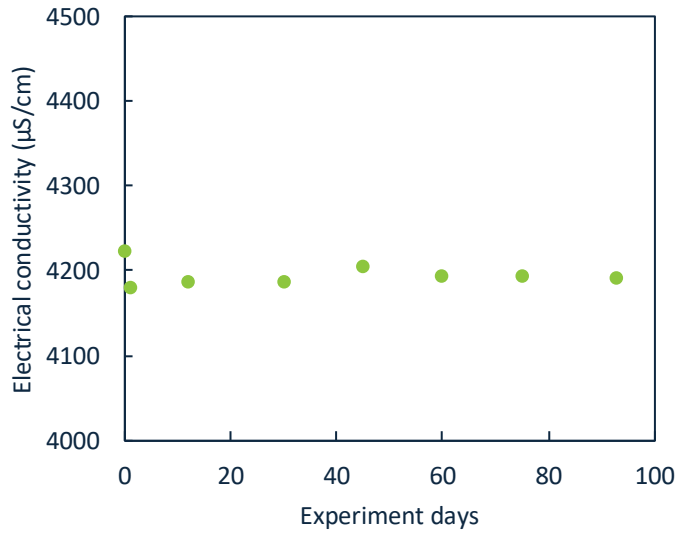


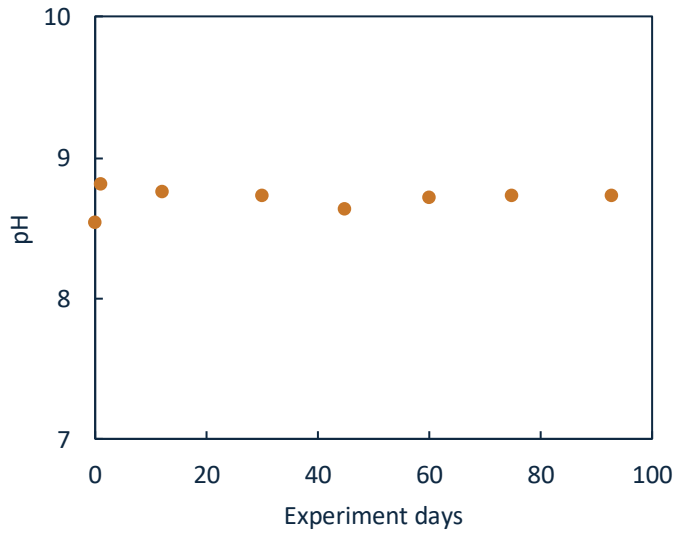


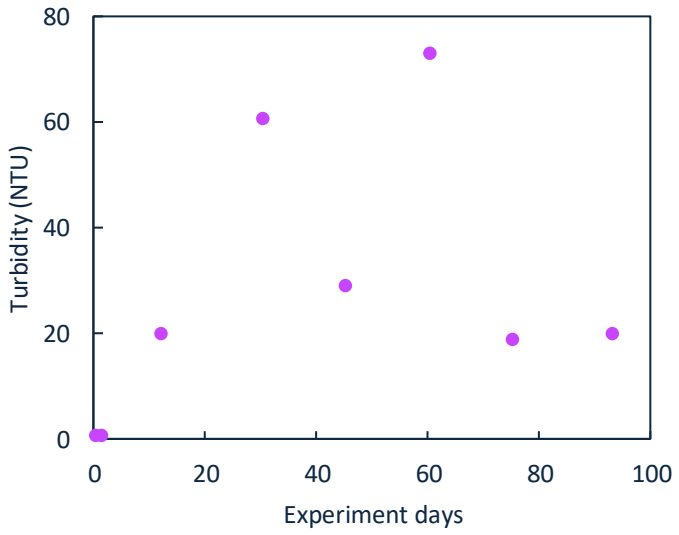
Wall thickness remained constant within variability range of ultrasonic transducer device after device change

Fluid sample analysis

Day		0	1	12	30	45	60	75	93
copper (ppm)	ASTM D6130	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1







- Copper element in fluid remained low, no significant change in electrical conductivity and pH
- Fluid samples developed haze over time and turbidity level increased

Analysis of particulates in fluids

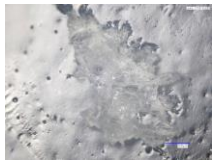
- Al and Si were detected from XRF
- Light microscope and SEM-EDS were performed to identify particulates



Floating particulates

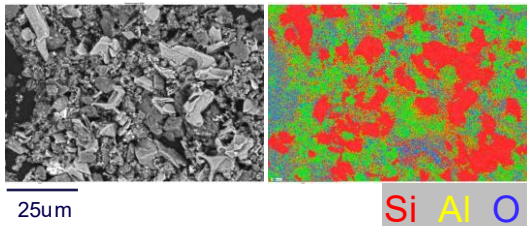
Sediments at the bottom

Light microscope



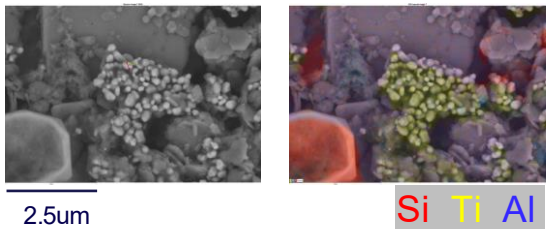
Possibly polymer

SEM-EDS



Possibly alumino-silicates/silicates

SEM-EDS

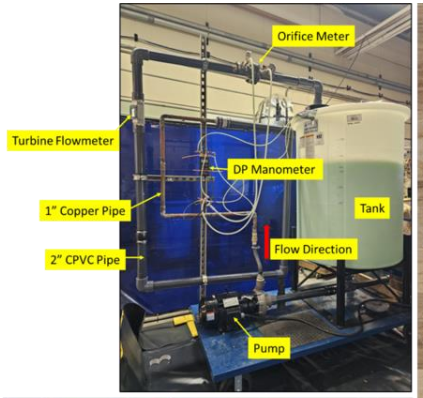


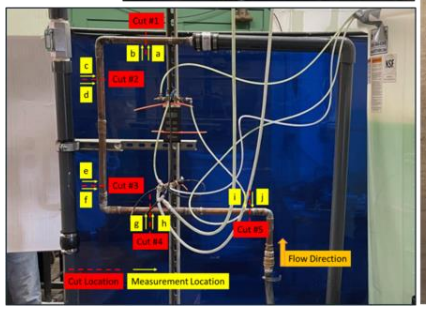
Possibly Titanium oxide

- Fluid samples were contaminated with aluminosilicates/silicates, titanium oxide, and polymer particulates during testing at the facility

Objective

- Determine the surface chemistry and layer structure protecting the copper, quantify any flow-driven differences between bent and straight sections, and flag risks (film thinning, sulfide/MIC, solder-metal dissolution) for prolonged data-center service.

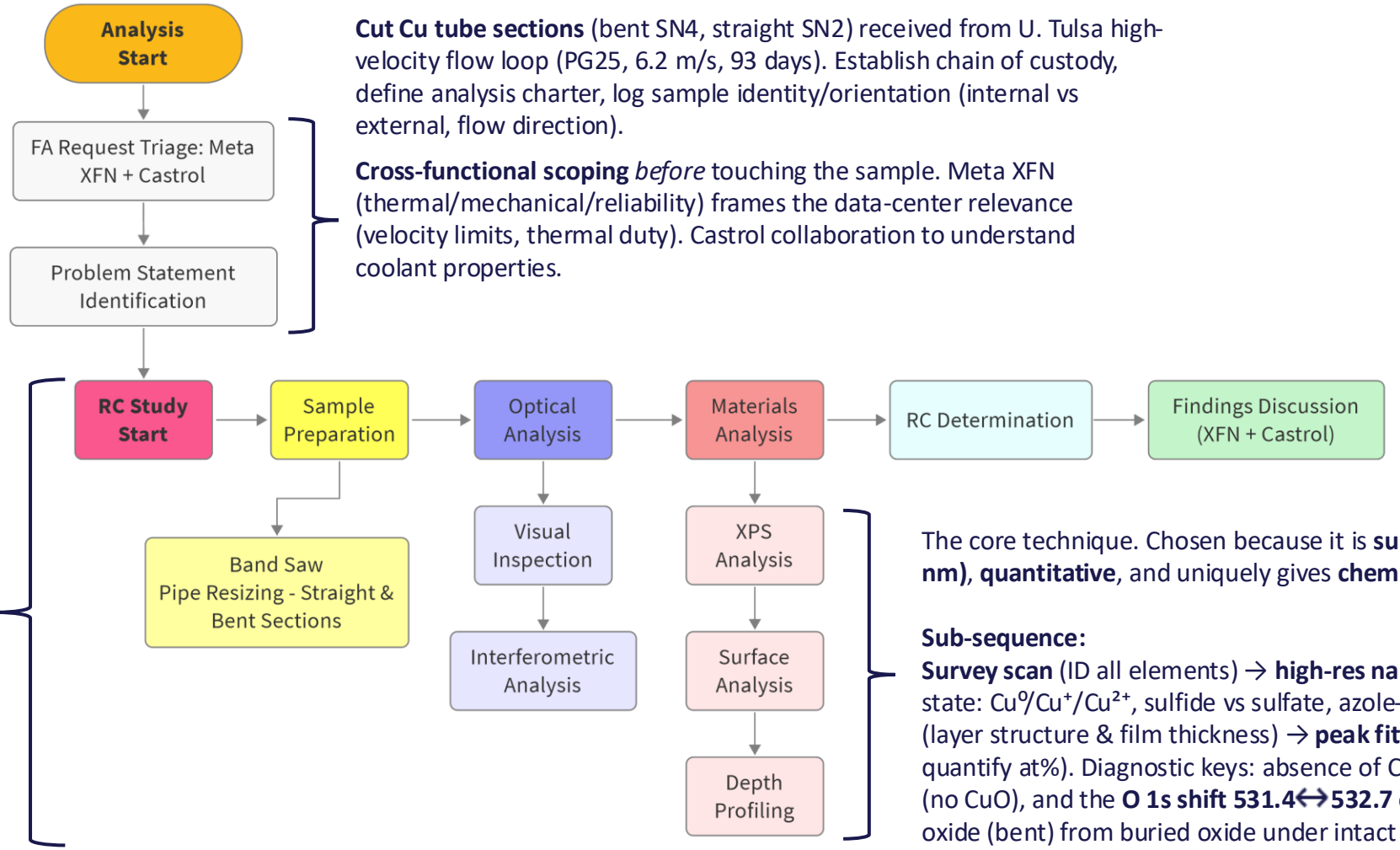








Failure Analysis Flow



Cut Cu tube sections (bent SN4, straight SN2) received from U. Tulsa high-velocity flow loop (PG25, 6.2 m/s, 93 days). Establish chain of custody, define analysis charter, log sample identity/orientation (internal vs external, flow direction).

Cross-functional scoping *before* touching the sample. Meta XFN (thermal/mechanical/reliability) frames the data-center relevance (velocity limits, thermal duty). Castrol collaboration to understand coolant properties.

Formal kickoff of the physical investigation with a defined hypothesis tree: (a) bulk erosion, (b) chemical corrosion, (c) inhibitor-film erosion, (d) MIC. Techniques are then sequenced **non-destructive → destructive** to preserve evidence.

The core technique. Chosen because it is **surface-sensitive (~5–10 nm)**, **quantitative**, and uniquely gives **chemical/oxidation state**.

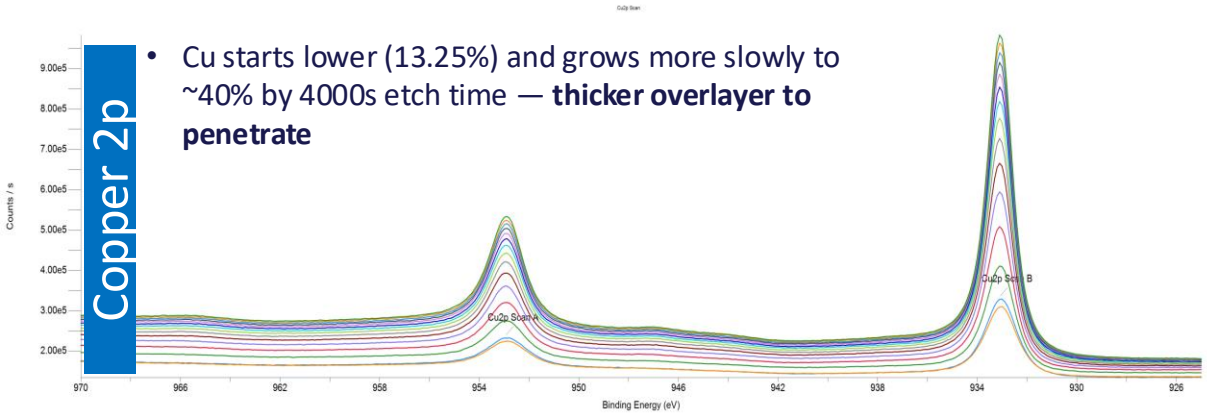
Sub-sequence:
Survey scan (ID all elements) → **high-res narrow scans** (chemical state: $Cu^0/Cu^+/Cu^{2+}$, sulfide vs sulfate, azole-N) → **Ar^+ depth profiling** (layer structure & film thickness) → **peak fitting** (deconvolve overlaps, quantify at%). Diagnostic keys: absence of Cu 2p shake-up satellites (no CuO), and the **O 1s shift 531.4↔532.7 eV** distinguishing exposed oxide (bent) from buried oxide under intact film (straight).

Cu Pipes – Bent Section



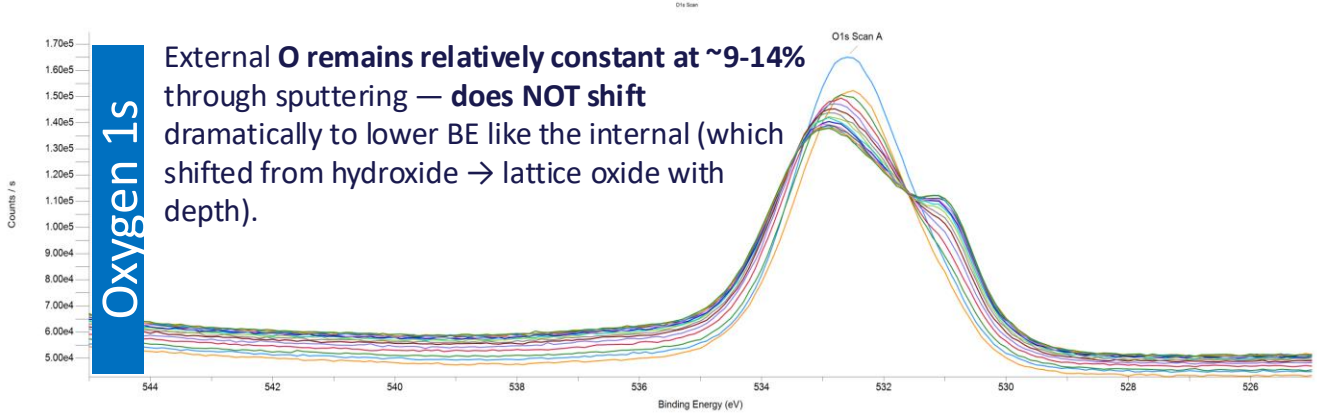


Copper 2p



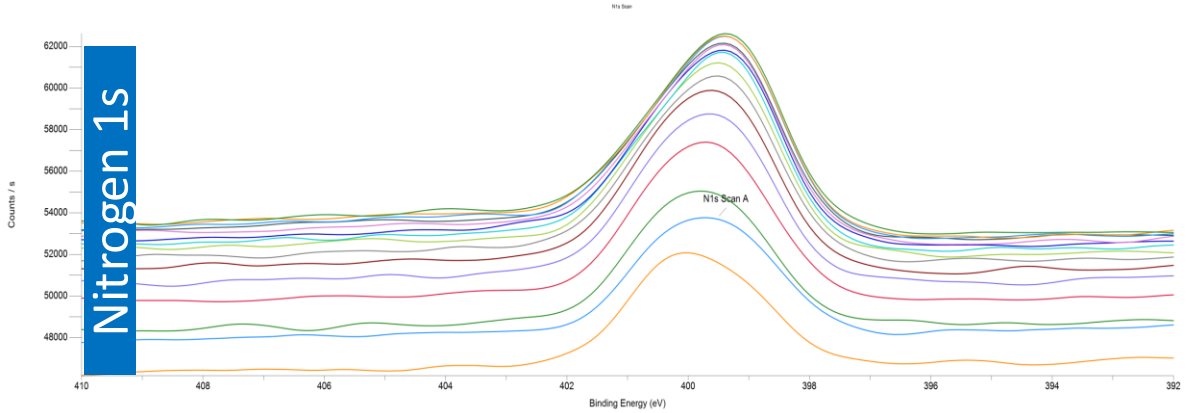
- Cu starts lower (13.25%) and grows more slowly to ~40% by 4000s etch time — **thicker overlayer to penetrate**

Oxygen 1s



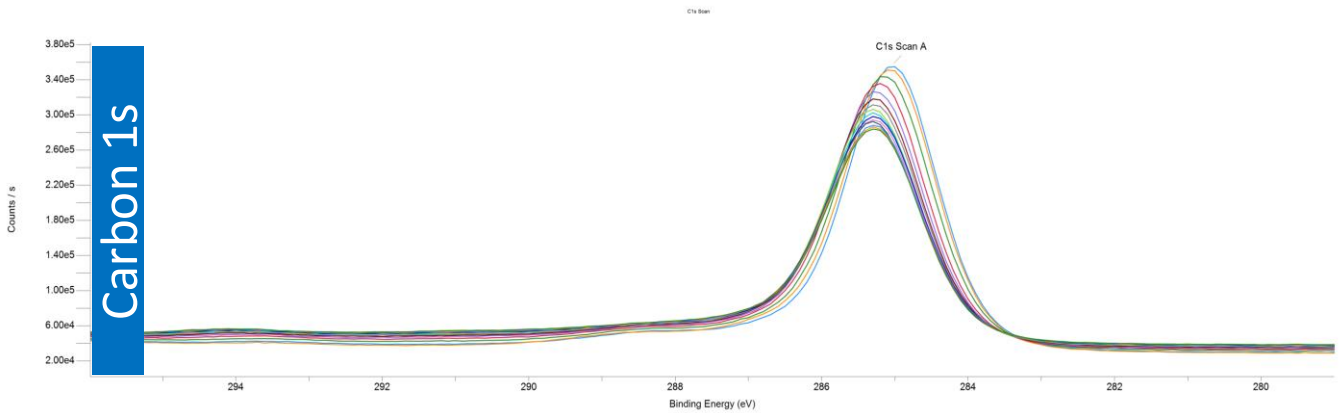
External **O remains relatively constant at ~9-14%** through sputtering — **does NOT shift** dramatically to lower BE like the internal (which shifted from hydroxide → lattice oxide with depth).

Nitrogen 1s



- **External N** grows with sputtering depth → environmental nitrogen incorporated throughout the contamination layer **(atmospheric organics, fingerprint residue).**

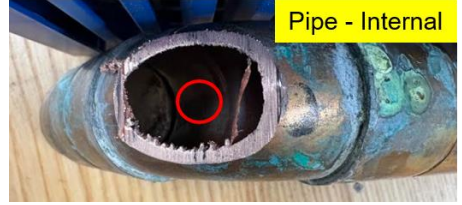
Carbon 1s

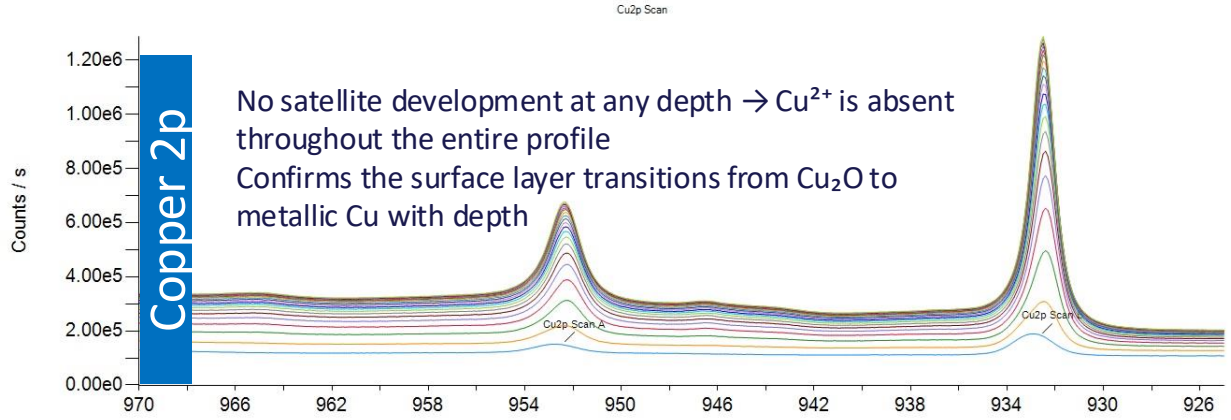


- **External C layer is much thicker** — persists at ~33% even after 4000s of sputtering (vs. internal where C decreases much faster).
- This thick carbon layer is acting as a **"protective barrier"** on the external surface, **preventing Cu^{2+} formation**.

Cu Pipes – Bent Section

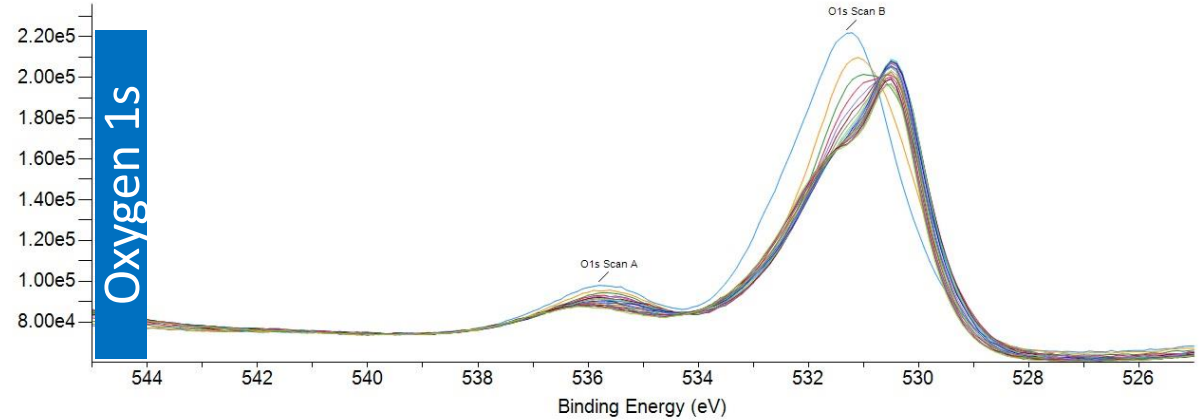




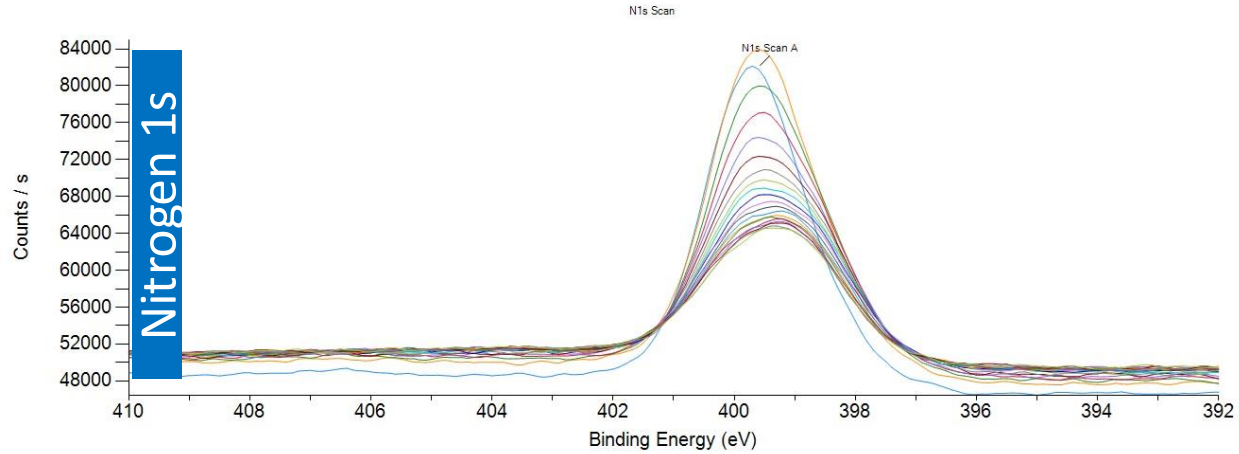


No satellite development at any depth → Cu^{2+} is absent throughout the entire profile

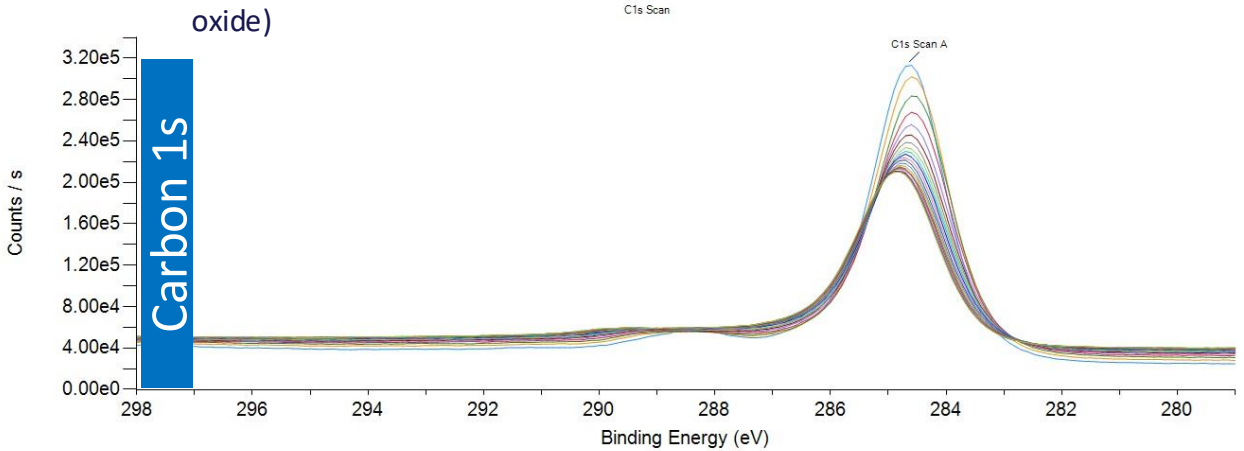
Confirms the surface layer transitions from Cu_2O to metallic Cu with depth



- Strong O at ~531 eV (hydroxide/organic O) + Na Auger at ~535.5 eV
- With depth: Peak shifts toward ~530 eV (from hydroxide → lattice oxide)



- **Azole protective film:** a monolayer-to-thin-film that forms at the Cu/PG25 interface
- **This confirms the corrosion inhibitor is working** — it's forming a surface-bound Cu–azole complex.



- Carbon is concentrated at the outermost surface layer
- **Confirms organic overlayer** (dispersant + corrosion inhibitor film). Gets sputtered away during depth profiling.

Cu Pipes – Bent Section





SN4 — Bent/Elbow Section (Internal)

Total Film: ~25–27 nm

SURFACE (Etch Level 0) — ~8 nm thick

- Organic overlayer: C-C/C-H + C-O (PG25 residue, **285.3 eV**)
  - C 1s = 46.83 at% (thinner — flow-eroded)
- BTA/TTA inhibitor film: N (=N–, **~399.8 eV**)
  - N 1s = 3.52 at% (thin remnant film)
- No Sn detected (no coupling at elbow)
- O 1s = 18.81 at% at **531.4 eV** (metal hydroxide — oxide exposed!)
- Cu clearly visible: **20.17 at%** (substrate showing through)

INTERMEDIATE (Etch Levels 1-10) — ~8 nm

- Sodium silicate (Na_2SiO_3): Na **~1072 eV**, Si **~102.8 eV**
  - Na = 6.28 at% (directly exposed — organic stripped)
- Zinc hydroxide ($Zn(OH)_2$): Zn **~1022 eV**, 0.61 at%
- Phosphate (PO_4^{3-}): P **~133 eV**
- Metal hydroxide: O **~531 eV**
- Cu_2S sulfide: S **~162 eV** (more prominent than straight)
- Lead deposit ($PbO/PbCO_3$): Pb **~138.5 eV**, 0.10 at%

SUBSURFACE OXIDE (Etch Levels 10-15) — ~5–7 nm

- Cu_2O (cuprous oxide): Cu **~932.5 eV**, O **~530 eV**
- Trace $Ni(OH)_2$: Ni **~855 eV**
- Na Auger visible in O 1s region (~535.5 eV)

SUBSTRATE (Etch Levels 15-19+) — reached at ~3000s

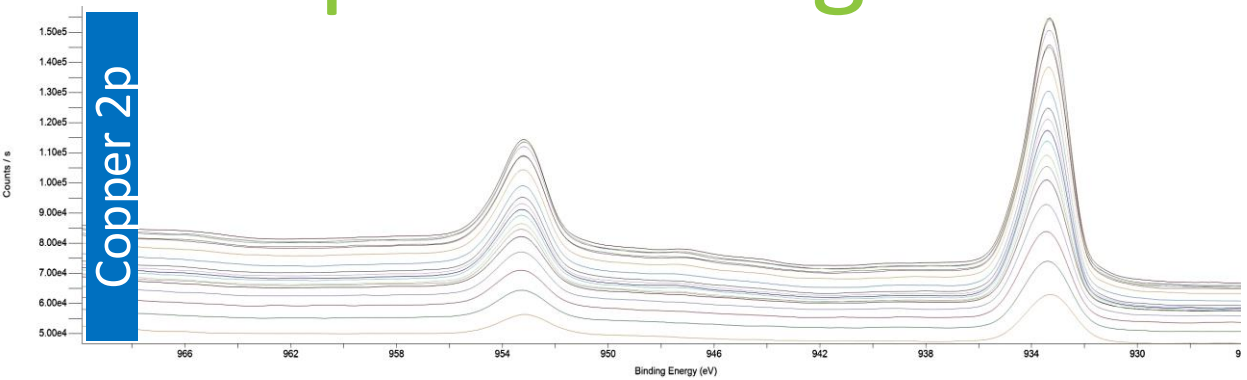
- Metallic Cu: Cu **~932.6 eV** (no satellites)
  - Cu becomes dominant >60% by Etch Level 15
- Trace Ni (alloy): residual Ni signal
- Residual Pb, Zn, Na (deep penetration/ion mixing)

- **Cu_2O –** Stable oxide layer noted in subsurface depth profiling. Dense, adherent, epitaxial layer which prevents further oxidation.
- **No Cu^{2+} (CuO)** — The absence of shake-up satellites in Cu 2p confirms no CuO (porous liner) at any depth. The copper is in Cu^0/Cu^+ states only, which is consistent with the mildly reducing environment inside a sealed glycol loop.
- **Corrosion inhibitor is present and functioning** — Azole (N 1s), sodium silicate (Na + Si), zinc (Zn^{2+}), and phosphate (P) all confirm an active inhibitor film.
- **Lead contamination (Pb)** — Pb is clearly identified with proper 4f doublet. Source is likely Pb solder dissolution. Even at 0.10 at%, this warrants attention.
- **Sulfide formation (Cu_2S)** — The dominant S 2p peak at 162 eV indicates sulfide attack on the Cu surface. This may indicate:
  - Galvanic or electrochemical sulfate reduction

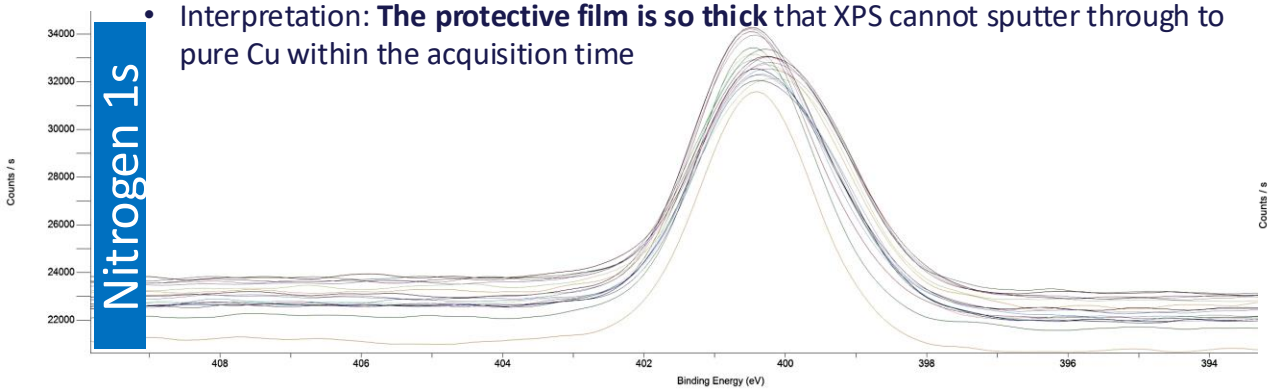
Cu Pipes – Straight Section



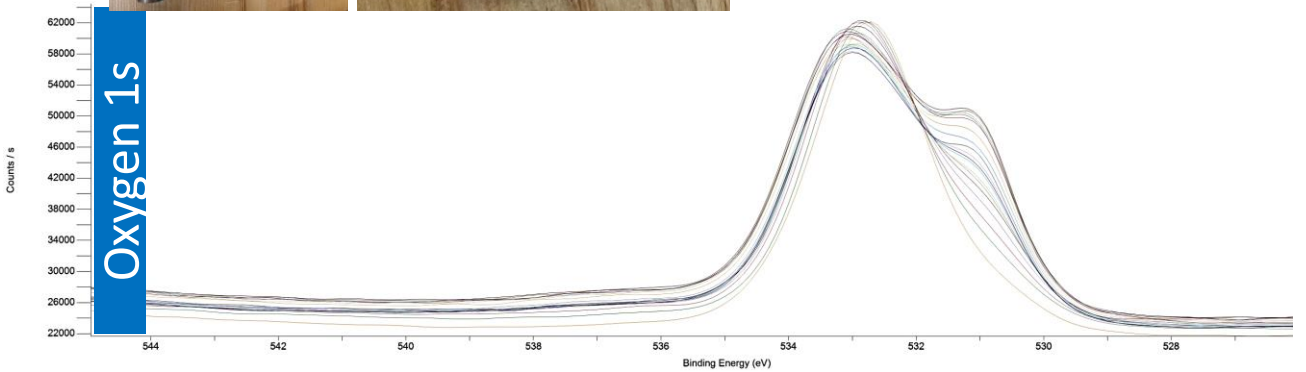




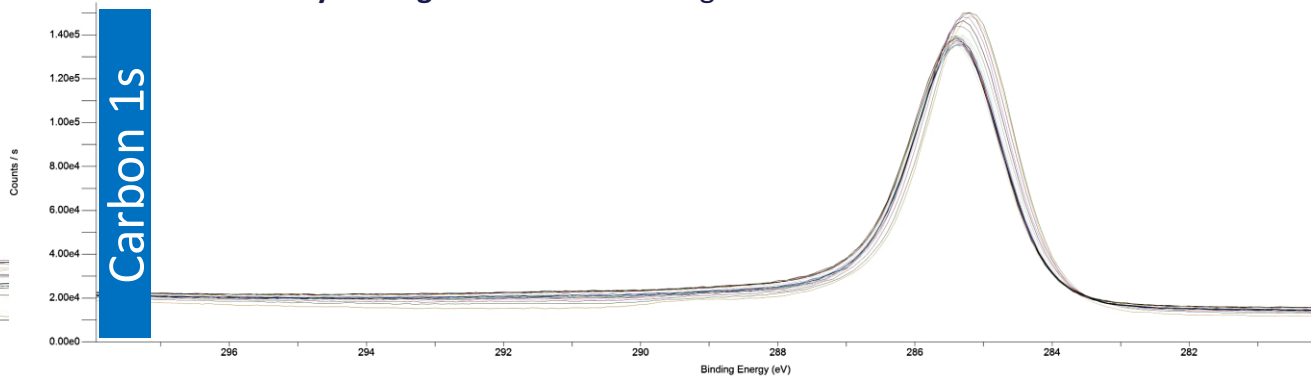
- **Cu grows slowly with sputtering** but never becomes the dominant signal even after 5600s
- **Maximum Cu reached: ~19 at%** (compared to >60% on bent section at same etch time)
- Interpretation: **The protective film is so thick** that XPS cannot sputter through to pure Cu within the acquisition time



- The 0.44 eV upshift vs. bent section (400.3 vs 399.8 eV) suggests more **"bulk" azole polymer (400+ eV) vs. thin Cu-coordinated monolayer (~399.5 eV)** on the bent section
- This is the strongest indicator that the **inhibitor film is fully intact** on the straight section



- The high BE indicates oxygen **is bonded to carbon** (ether/hydroxyl from PG25) and silicon (SiO_2/silicate)
- This 1.25 eV shift is direct chemical evidence that the **Cu_2O layer is completely covered by the organic film** on the straight section



- 64.21 at% — significantly higher than bent section (46.83 at%). This is the thickest organic overlayer of all surfaces analyzed.
- The high carbon content confirms the **Azole + PG25 organic film is intact** and undisturbed by the 6.2 m/s flow in the straight section

Cu Pipes – Straight Section





SN2 — Straight Section (Internal)

Total Film: ~44–46 nm

SURFACE (Etch Level 0) — ~35 nm thick

- Organic overlayer: C-C/C-H + C-O (PG25 residue, **285.3 eV**)
  - C 1s = 64.21 at% (dominant, thick film)
- BTA/TTA inhibitor film: N (=N–, **~400.3 eV**)
  - N 1s = 8.66 at% (2.5× thicker than bent)
- SnO_2 from Type D coupling solder (**~487 eV**)
- O 1s = 18.29 at% at **532.7 eV** (organic C-O, NOT metal oxide)
- Cu barely visible: **only 2.11 at%** (buried under organic)

INTERMEDIATE (Etch Levels 1-14+) — ~4 nm

- Sodium silicate (Na_2SiO_3): Na **~1072 eV**, Si **~103 eV**
  - Na = 1.11 at% (attenuated by thick organic above)
- Zinc hydroxide ($Zn(OH)_2$): Zn **~1023 eV**, 0.50 at%
- Phosphate (PO_4^{3-}): P **~133 eV**
- Cu_2S sulfide: S **~162 eV** (trace)
- Lead deposit ($PbO/PbCO_3$): Pb **~139 eV**, 0.10 at%

SUBSURFACE OXIDE — ~3–5 nm (estimated, not reached)

- Cu_2O (cuprous oxide): Cu **~933 eV**, O **~530 eV**
  - Estimated — not reached in 5600s sputtering
- Ni trace (from alloy): residual signal

SUBSTRATE (NOT reached in 5600s sputtering)

- Metallic Cu: Cu **~932.6 eV** (no satellites)
  - ⚠ Cu never becomes dominant — film intact
- C still >40% at maximum etch depth

- **Thick Azole/organic passivation film (~35 nm)** — The straight section retains a substantially thicker protective organic/azole film compared to the bent section.
- The **Cu_2O** passivating oxide is confirmed from the Cu^0/Cu^+ oxidation state (933.2 eV, no satellites). This confirms the protective film is intact and undisturbed by flow.
- The absence of **shake-up satellites** in Cu 2p confirms no CuO at any depth.
- **Corrosion inhibitor film is thicker and better-preserved than at the elbow** — Azole (N 1s = 8.66 at%, 2.5× more than bent section), sodium silicate (Na 1s = 1.11 at%, attenuated by thick organic), zinc (Zn^{2+} = 0.50 at%), and phosphate (P) are all present.

Lead contamination (Pb) – Same as SN4. Deposited throughout the loop regardless of flow geometry.

Sulfide (Cu_2S) less prominent — S 2p shows the same sulfate + sulfide pattern as SN4, but the sulfide signal is weaker and closer to noise level on the straight section. This is consistent with:

- The thicker organic/azole film physically blocking sulfide access to the Cu surface.
- Less turbulent flow in the straight section creating fewer anaerobic micro-environments.
- Sulfide accumulation preferentially occurring where the protective film is thinnest (i.e., the elbow).

Conclusions

- **No bulk metal erosion at 6.2 m/s over 93 days.**
- **Copper is fully passivated as Cu_2O (Cu^+) — no aggressive oxidation anywhere.**
- **A multi-layer protective film is present and functioning.**
  - Soft outer azole/organic film,
  - Hard inorganic Na-silicate + $Zn(OH)_2$ + phosphate layer,
  - Cu_2O oxide — confirmed by their BE (N 1s ~400 eV, Na/Si silicate, Zn^{2+}, PO_4^{3-} at 133 eV).
- **Flow-accelerated inhibitor film thinning occurs at bends — the central finding.**
  - The soft azole/organic film is **4.4× thinner at the elbow (~8 nm) than the straight section (~35 nm)**.
  - Turbulent shear at the bend strips the soft polymeric layer while leaving the hard inorganic layer beneath intact. The O 1s shift (531.4 eV exposed oxide at bend vs. 532.7 eV buried oxide on straight) is the direct chemical proof.
- **Sulfide (Cu_2S) forms preferentially where the film is thinnest.**
  - S^{2-} at 162 eV is more prominent on the thinned bent section, indicating sulfide attack on exposed copper — a potential microbiologically-influenced-corrosion (MIC) indicator. The pipes were not passivated prior to testing, and sulfide seems to be introduced from the pipes.
- **PG25 is safe at 6.2 m/s, but bends carry a reduced protection margin requiring monitoring, especially if coolants get contaminated.**
  - The system works as designed today, but continuous azole stripping at elbows could eventually outpace replenishment **if coolant contamination occurs**. Monitoring azole reserve critical, BART SRB testing for MIC, and adoption of Pb-free brazing.

Acknowledgements

- Castrol Team Members
  - Luis Alberto Castrillo (Research technologist)
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  - Prof. Siamack A. Shriazi
- Meta team members
  - John Fernandez (Thermal Engineer)
  - Cheng Chen (Thermal Engineer)
  - Jia Li (Failure Analysis Engineer)
  - Nikesh Tayi Dhar (Head of Hardware Product Integrity)

Call to Action

- Let’s collaborate on how to further test erosion
- Evaluate additional velocities, temperatures, geometries, materials, fluid formulations, and exposure durations
- Collaborate on repeatable methods, acceptance criteria, and technically justified design recommendations.

COOLING:
COLD PLATE



Evaluating material erosion in liquid cooling components – expedited screening

Greg Spangler

Sr Mgr CDP & Technical Leader – SABIC Specialty Resins

Content

1) Premise
2) Technical approach
3) Testing
4) Results
5) Conclusion

Premise

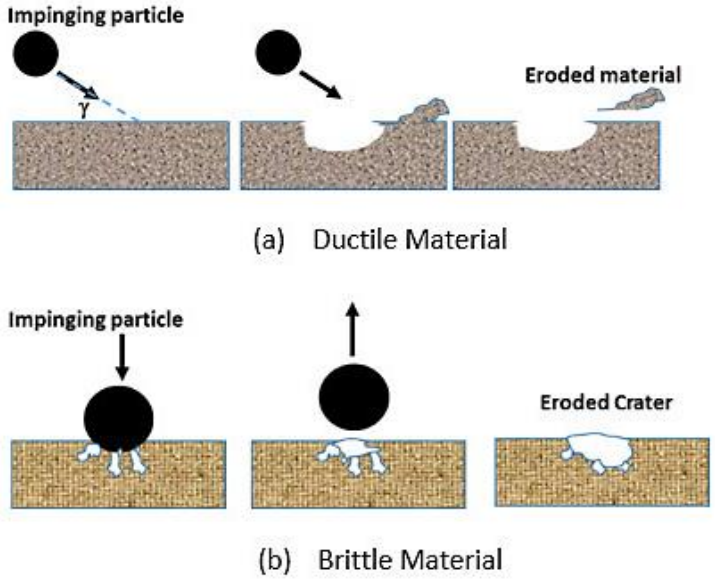
- Building on 2025 OCP Presentation – *PG25 High Velocity Test*

1) Two options for higher thermal design power (TDP) requirements
   - Reduce coolant temperature
   - Increase flow rate by raising fluid velocity (energy efficient)
2) Investigation into pipe erosion rates
   - Erosion a limiting factor in designs
   - Testing is lengthy process: need for improvements
3) Challenges
   - Can field be narrowed with better screening?
   - Alternative materials like thermoplastics offer advantages?

Technical Approach

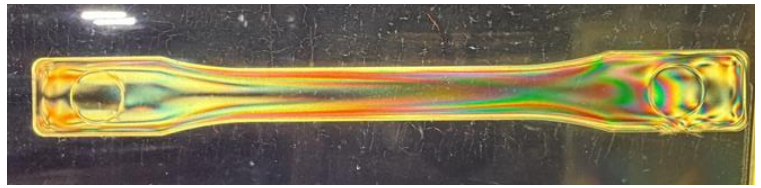
Physical (mechanical)

- Erosion rate proportional to particle concentration, impact angle, and impact velocity*
- With typical velocity ~5 m/s – not enough energy to cause erosion



Chemical

- Corrosion in metals results in material change and local failure
- Thermoplastics have molded in stress; ESCR potential mode of failure



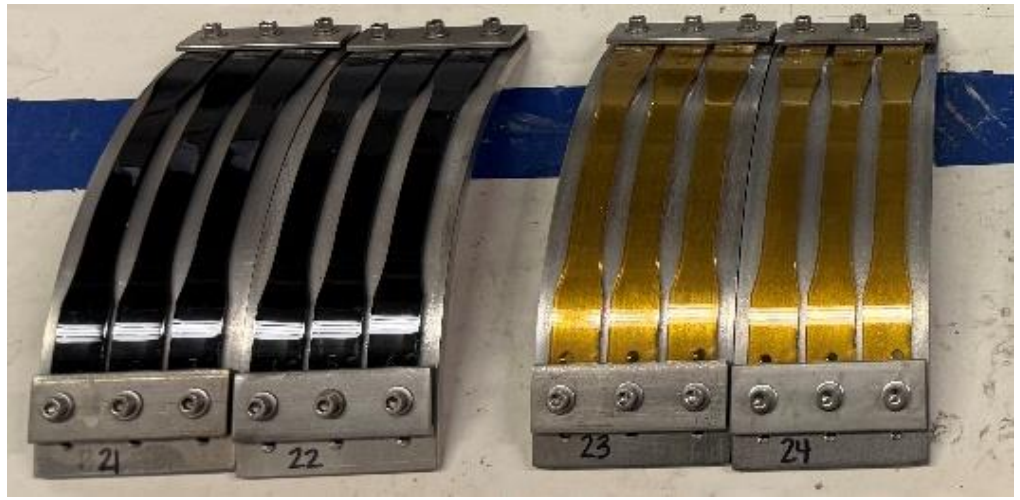
Birefringence image: qualitatively molded-in stress:

* See references page 8

Test Method

Environmental stress cracking resistance test – ASTM D543 immersion in PG25

- Evaluation criteria
  - Visual
  - Dimensional change
  - Weight change
  - Mechanical property retention



- Initial Screening
  - Unstrained (0%), strained (0.5%, 1.0%) in air and immersed – 23deg C for 7 days
- Ongoing Expansion
  - Unstrained (0%), strained (1%) in hot air and immersed – 80deg C for 14 days and 42 days

Conclusion

Work is ongoing…

- Results to be added and tabulated upon completion of each condition
- Further refine evaluation criteria
  - Correlation of accelerated conditions to performance testing
  - Measurement criteria
- Contributing factors
  - Inhibitors, additive and contaminant impact, beyond PG25
  - Flow or circulation
  - Adaptation of ASTM D543 for liquid cooling applications

[Call to Action]

- Problem to solve
  - A robust methodology for accelerated screening of long-term material performance in liquid cooling applications – at diverse conditions and beyond metals
- Currently a study to develop a proposal but encourage input and commentary
- Correlate to test methods used for other material families
- Initial screening on first round material complete 1Q2027
- Where to find additional/reference information (URL links)

Theoretical Prediction Method for Erosion Damage of Horizontal Pipe by Suspended Particles in Liquid–Solid Flows, Materials 2021

Liquid–Solid Impact Mechanism, Liquid Impingement Erosion, and Erosion-Resistant Surface Engineering: A Review, Coatings 2023

Better Particle Erosion Fluid Dynamics Modeling in Ansys Fluent | Ansys, ANSYS Blog 2017

Guidelines for Using Water-Based Transfer Fluids in Single-Phase Cold Plate-Based Liquid-Cooled Racks, OCP

Thank You!

